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(54) **GENERATION OF AN OPTICAL
LOCAL-OSCILLATOR SIGNAL FOR A
COHERENT-DETECTION SCHEME**

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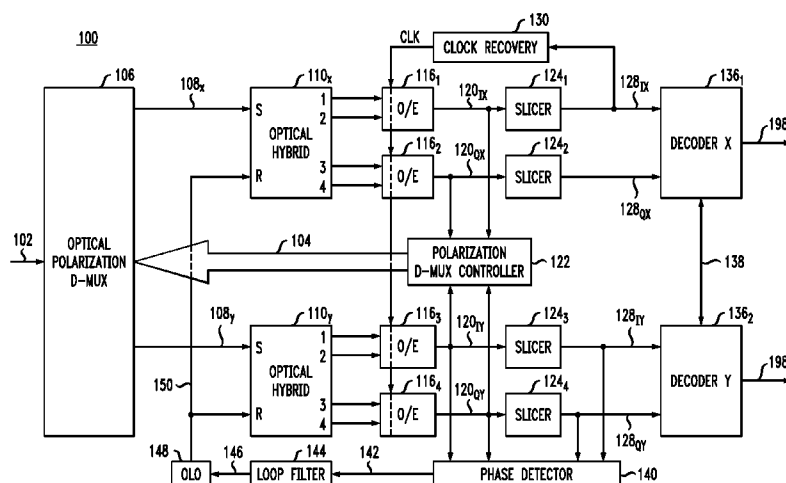
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(57) **ABSTRACT**

An optical receiver having an optical IQ modulator config-
ured to generate an optical local-oscillator (OLO) signal for
optical homodyne detection of an optical input signal applied
to the optical receiver. The optical receiver further has (i) a
phase detector configured to generate an electrical measure of
the phase difference between the OLO signal and a carrier
wave of the optical input signal and (ii) a phase-lock loop
configured to drive the optical IQ modulator using the elec-
trical measure. In an embodiment, the phase detector is con-
figured to generate the electrical measure using both I and Q
components of the homodyne-detected signal and in a man-
ner that enables the optical receiver to be compatible with the
M-QAM modulation format.

16 Claims, 5 Drawing Sheets



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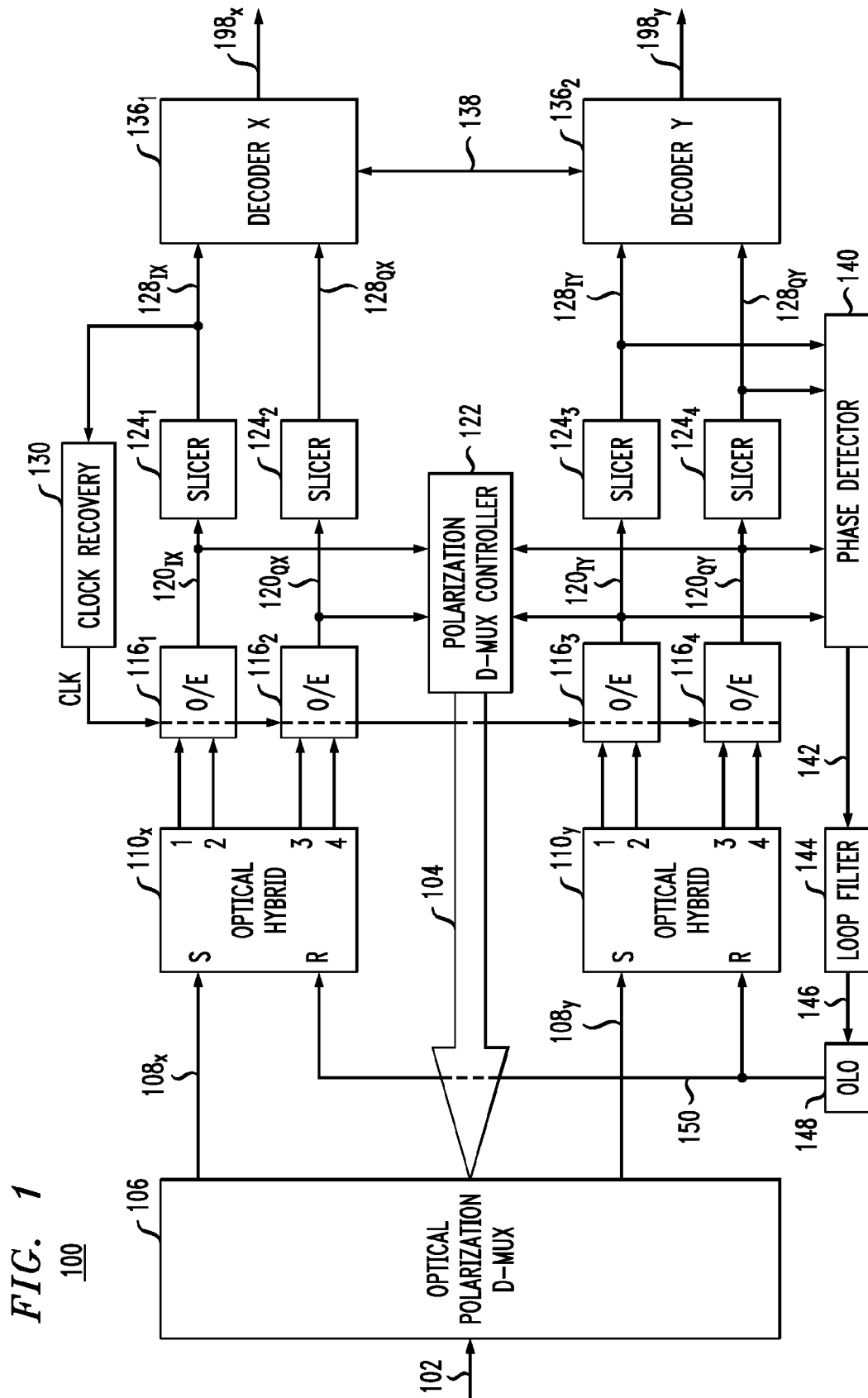
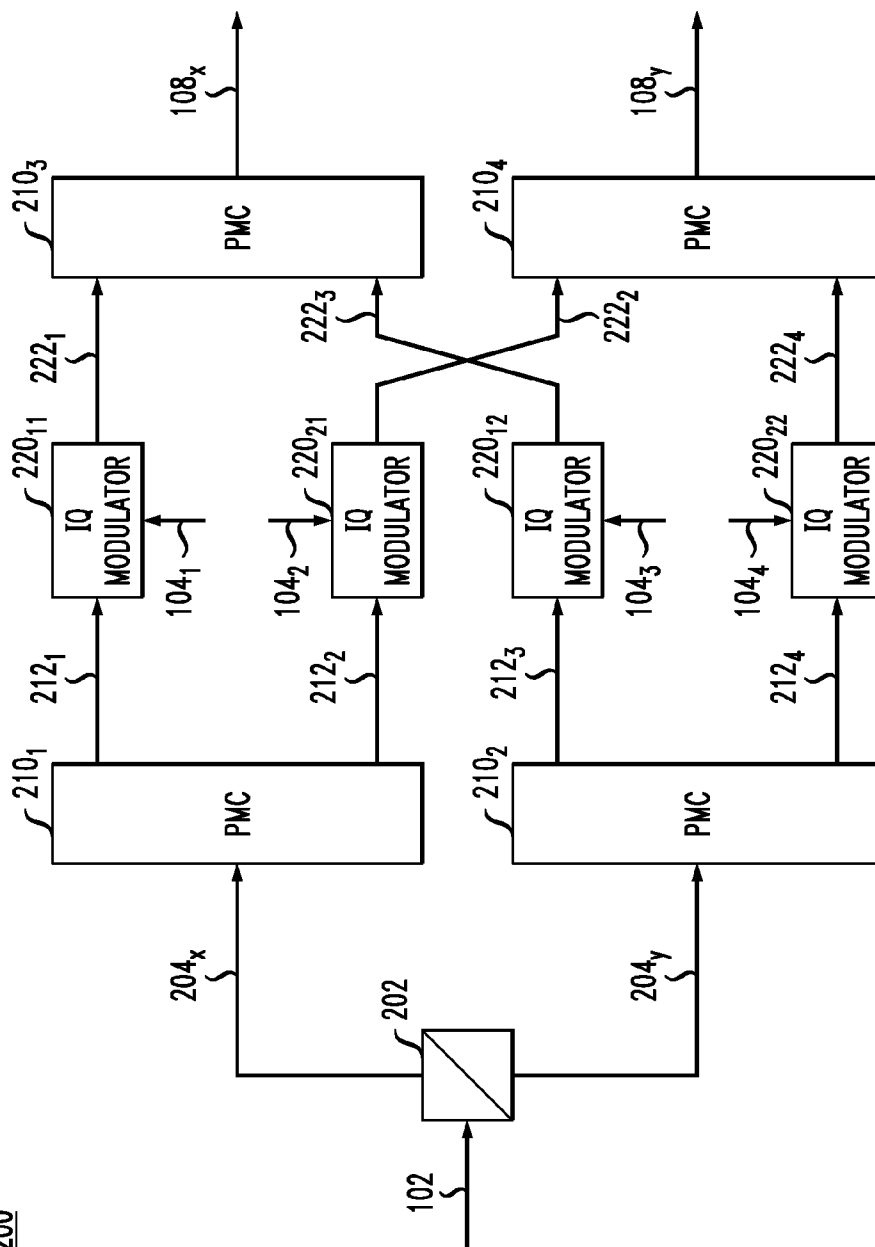


FIG. 2

200



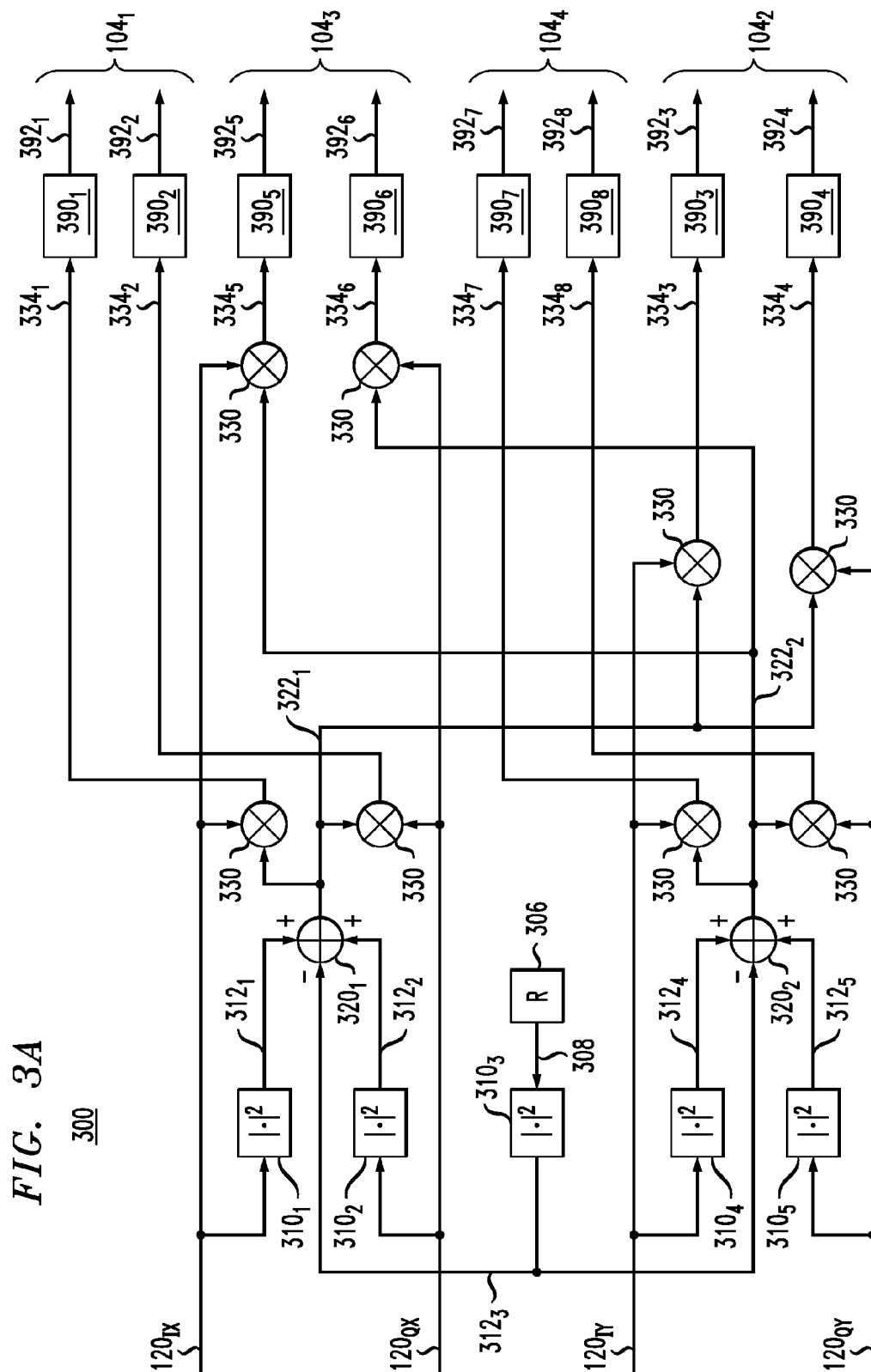


FIG. 3B

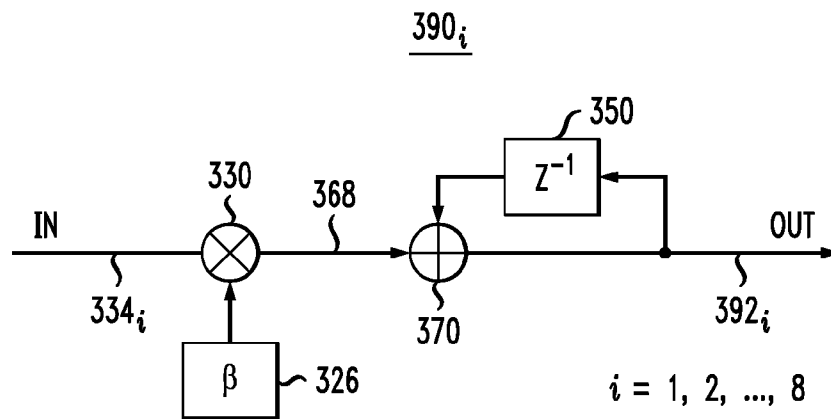


FIG. 4

400

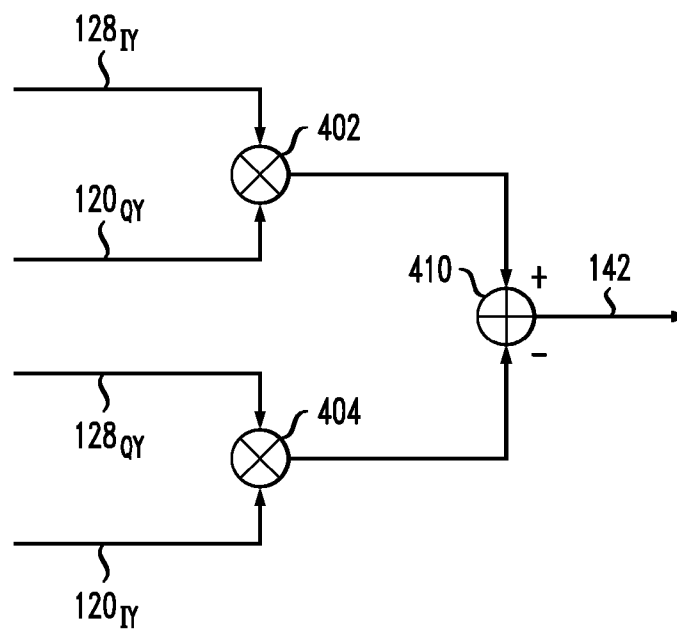


FIG. 5

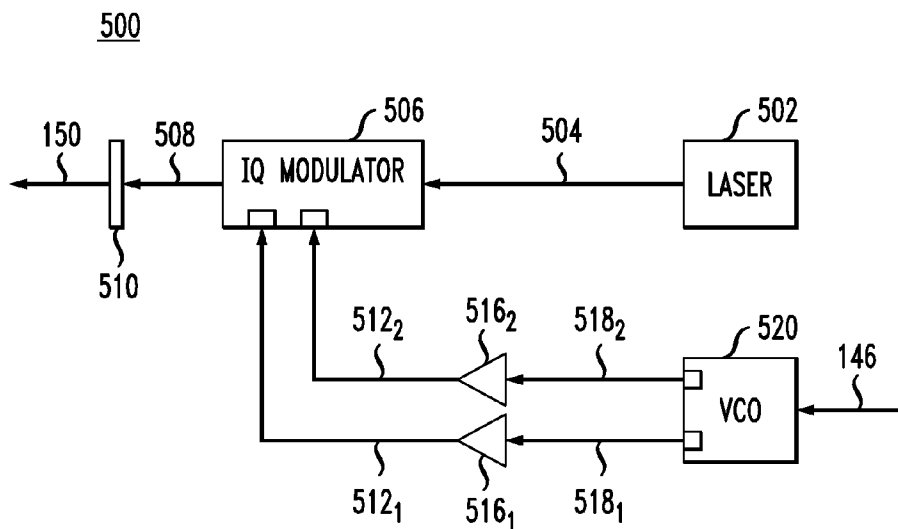
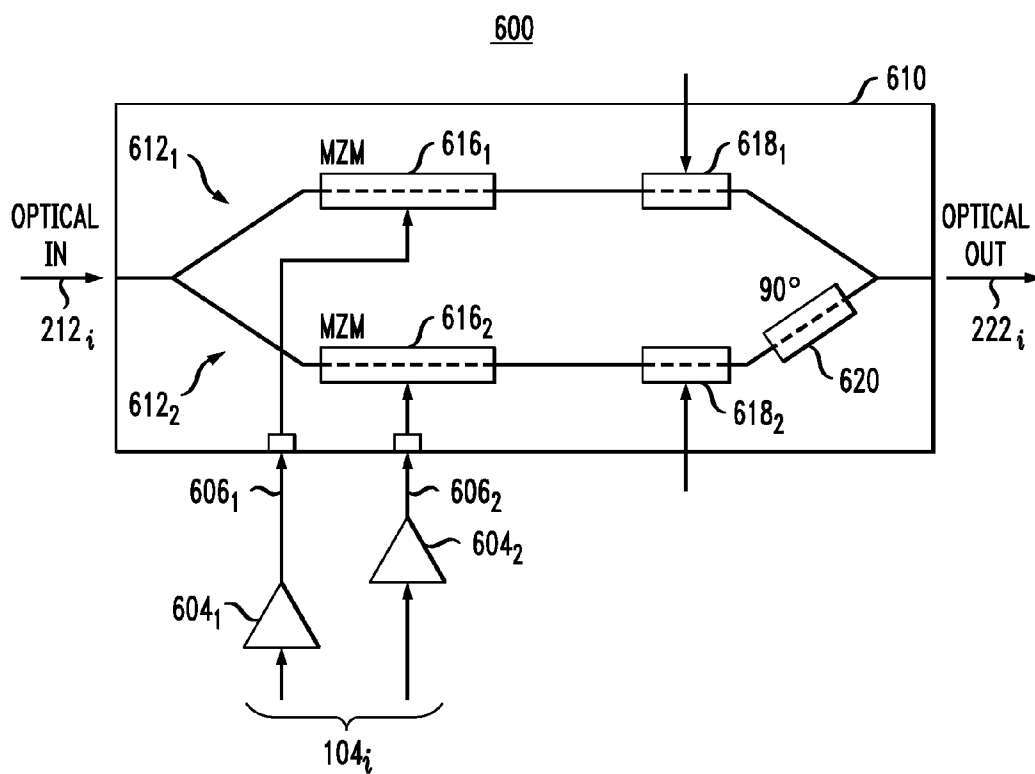


FIG. 6



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GENERATION OF AN OPTICAL LOCAL-OSCILLATOR SIGNAL FOR A COHERENT-DETECTION SCHEME

CROSS-REFERENCE TO RELATED APPLICATIONS

The subject matter of this application is related to that of U.S. Patent Application Publication No. 2014/0199068, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

The present disclosure relates to optical communication equipment and, more specifically but not exclusively, to coherent detection of modulated optical signals.

2. Description of the Related Art

This section introduces aspects that may help facilitate a better understanding of the invention(s). Accordingly, the statements of this section are to be read in this light and are not to be understood as admissions about what is in the prior art or what is not in the prior art.

Coherent detection is used in optical transport systems, for example, because it is compatible with advanced modulation formats, such as quadrature phase-shift keying (QPSK) and quadrature-amplitude modulation (QAM), which enable the corresponding system to achieve a relatively high spectral efficiency, e.g., better than 1 bit/s/Hz. The use of polarization multiplexing serves to further increase the bit rate and spectral efficiency approximately by a factor of two. However, a major challenge of this particular transmission format is the need for fast and precise polarization tracking and, in the case of homodyne detection, for phase and carrier-frequency locking of the optical local-oscillator signal.

SUMMARY OF SOME SPECIFIC EMBODIMENTS

Disclosed herein are various embodiments of an optical receiver having an optical IQ modulator configured to generate an optical local-oscillator (OLO) signal for optical homodyne detection of an optical input signal applied to the optical receiver. The optical receiver further has (i) a phase detector configured to generate an electrical measure of the phase difference between the OLO signal and a carrier wave of the optical input signal and (ii) a phase-lock loop configured to drive the optical IQ modulator using the electrical measure. In an embodiment, the phase detector is configured to generate the electrical measure using both I and Q components of the homodyne-detected signal and in a manner that enables the optical receiver to be compatible with the M-QAM modulation format.

According to one embodiment, provided is an apparatus comprising: an optical hybrid configured to optically mix an optical input signal and an OLO signal to generate a plurality of mixed optical signals; a first optical-to-electrical converter configured to convert a first subset of the plurality of mixed optical signals into a first electrical analog signal (e.g., **120_I**); a second optical-to-electrical converter configured to convert a second subset of the plurality of mixed optical signals into a second electrical analog signal (e.g., **120_Q**); a first slicer circuit configured to slice the first electrical analog signal to generate a first electrical digital signal (e.g., **128_I**); a second slicer circuit configured to slice the second electrical analog signal to generate a second electrical digital signal (e.g., **128_Q**); and a phase detector configured to generate an elec-

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trical measure of a phase difference between the OLO signal and a carrier wave of the optical input signal, with said electrical measure being generated using the first electrical analog signal, the first electrical digital signal, the second electrical analog signal, and the second electrical digital signal.

In some embodiments of the above apparatus, the apparatus further comprises a decoder configured to recover data carried by the optical input signal from the first electrical digital signal and the second electrical digital signal.

In some embodiments of any of the above apparatus, the apparatus is configured to use said electrical measure to phase-lock the OLO signal to the carrier wave of the optical input signal.

In some embodiments of any of the above apparatus, said electrical measure is proportional to the following quantity: $I_y Q_y - Q_y' I_y$, where I_y , Q_y , Q_y' , and I_y' are magnitudes of the first electrical digital signal, the second electrical analog signal, the second electrical digital signal, and the first electrical analog signal, respectively.

In some embodiments of any of the above apparatus, the phase detector comprises: a first signal multiplier configured to multiply the first electrical analog signal and the first electrical digital signal to generate a first multiplied electrical signal; a second signal multiplier configured to multiply the second electrical analog signal and the second electrical digital signal to generate a second multiplied electrical signal; and a signal adder configured to subtract the second multiplied electrical signal from the first multiplied electrical signal to generate the electrical measure.

In some embodiments of any of the above apparatus, the apparatus further comprises: a laser configured to generate an optical beam; and an optical IQ modulator configured to optically modulate said optical beam to generate a modulated optical beam, wherein the apparatus is configured to generate the OLO signal using said modulated optical beam.

In some embodiments of any of the above apparatus, the optical IQ modulator is configured to optically modulate the optical beam using carrier-suppression and sideband-suppression modulation.

In some embodiments of any of the above apparatus, the apparatus further comprises an optical filter configured to filter the modulated optical beam to generate the OLO signal.

In some embodiments of any of the above apparatus, the optical filter is configured to pass a first modulation sideband of the modulated optical beam; and the OLO signal comprises light of the first modulation sideband that passed through the optical filter.

In some embodiments of any of the above apparatus, the optical filter is further configured to block a carrier wave and a second modulation sideband of the modulated optical beam.

In some embodiments of any of the above apparatus, the apparatus further comprises a phase-lock loop configured to drive the optical IQ modulator using the electrical measure.

In some embodiments of any of the above apparatus, the phase-lock loop comprises: an electrical loop filter configured to filter the electrical measure to generate a filtered measure; and an electrical voltage-controlled oscillator configured to drive the optical IQ modulator using the filtered measure.

In some embodiments of any of the above apparatus, the electrical voltage-controlled oscillator is configured to: generate a first electrical radio-frequency signal using the filtered measure; and generate a second electrical radio-frequency signal using the filtered measure, wherein the first and second electrical radio-frequency signals have a common radio-frequency proportional to the filtered measure, and there is a fixed phase offset of approximately 90 degrees between the

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first electrical radio-frequency signal and the second electrical radio-frequency signal. The phase-lock loop is configured to drive the optical IQ modulator using the first and second electrical radio-frequency signals.

In some embodiments of any of the above apparatus, the IQ modulator comprises a nested Mach-Zehnder modulator.

In some embodiments of any of the above apparatus, the apparatus further comprises an optical polarization de-multiplexer configured to de-multiplex an optical polarization-division-multiplexed signal into a first optical component having a first polarization and a second optical component having a second polarization orthogonal to the first polarization, wherein the optical input signal is one of said first and second optical components.

In some embodiments of any of the above apparatus, the apparatus further comprises a controller configured to generate one or more control signals for the optical polarization de-multiplexer using the first electrical analog signal and the second electrical analog signal.

According to another embodiment, provided is an apparatus comprising: an optical IQ modulator configured to optically modulate a coherent optical beam to generate a modulated optical beam, wherein the apparatus is configured to generate an OLO signal using said modulated optical beam; an opto-electric circuit configured to apply optical homodyne detection, using the OLO signal, to an optical input signal; and a phase detector configured to generate an electrical measure of a phase difference between the OLO signal and a carrier wave of the optical input signal, wherein the apparatus is configured to drive the optical IQ modulator using the electrical measure.

In some embodiments of the above apparatus, the apparatus further comprises: an electrical loop filter configured to filter the electrical measure to generate a filtered measure; and an electrical voltage-controlled oscillator configured to drive the optical IQ modulator using the filtered measure.

In some embodiments of any of the above apparatus, the optical IQ modulator is further configured to optically modulate the coherent optical beam using at least one of carrier-suppression and sideband-suppression modulation.

In some embodiments of any of the above apparatus, the apparatus further comprises an optical filter configured to filter the modulated optical beam to generate the OLO signal, wherein: the optical filter is configured to pass a first modulation sideband of the modulated optical beam; and the OLO signal comprises light of the first modulation sideband that passed through the optical filter.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments of the invention will become more fully apparent from the following detailed description and the accompanying drawings, in which:

FIG. 1 shows a block diagram of a coherent receiver according to an embodiment of the disclosure;

FIG. 2 shows a block diagram of an optical polarization de-multiplexer that can be used in the coherent receiver shown in FIG. 1 according to an embodiment of the disclosure;

FIGS. 3A-3B show block diagrams of a polarization controller that can be used in the coherent receiver shown in FIG. 1 according to an embodiment of the disclosure;

FIG. 4 shows a block diagram of a phase detector that can be used in the coherent receiver shown in FIG. 1 according to an embodiment of the disclosure;

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FIG. 5 shows a block diagram of a light source that can be used in the coherent receiver shown in FIG. 1 according to an embodiment of the disclosure; and

FIG. 6 shows a block diagram of an IQ modulator that can be used in the optical polarization de-multiplexer shown in FIG. 2 according to an embodiment of the disclosure.

DETAILED DESCRIPTION

FIG. 1 shows a block diagram of a coherent receiver 100 according to an embodiment of the disclosure. Receiver 100 is configured to receive a polarization-division-multiplexed (PDM) optical input signal 102, e.g., from a remote transmitter (not explicitly shown in FIG. 1), and process the received signal, e.g., as further described below, to recover the data carried by that signal. Receiver 100 then directs the recovered data, via output data streams 198x and 198y, to the intended data destination(s). In an alternative embodiment, receiver 100 may include a multiplexer (not explicitly shown in FIG. 1) configured to combine data streams 198x and 198y into a corresponding single output data stream.

In one embodiment, optical input signal 102 is delivered to receiver 100 via an optical fiber (not explicitly shown in FIG. 1). Different sections of the optical fiber typically have different principal states of polarization (PSPs) that are not necessarily aligned with the original (e.g., independently modulated) PDM components of signal 102. This misalignment causes each of the PDM components to generally have a non-zero projection on each of the PSPs, which mixes the PDM components and alters the polarization state of the optical signal. In addition, the PSPs of the optical fiber tend to change over time, e.g., due to varying physical conditions along the length of the fiber. This temporal variation causes the polarization state of optical input signal 102 to vary over time as well.

An optical polarization de-multiplexer (D-MUX) 106 serves to undo, to a significant extent, the polarization rotation imposed onto optical input signal 102 by the optical fiber. More specifically, polarization D-MUX 106 is configured to de-multiplex optical input signal 102 into optical signals 108x and 108y in a manner that causes each of optical signals 108x and 108y to represent a respective one of the original PDM components generated at the remote transmitter with as little crosstalk from the other PDM component as practically possible. The configuration of polarization D-MUX 106 is controllable via a multi-component control signal 104 generated by a controller 122. By dynamically changing control signal 104, controller 122 can configure polarization D-MUX 106 to appropriately track the polarization-rotation changes in the optical fiber, thereby enabling receiver 100 to detect and decode optical signals 108x and 108y in a relatively straightforward manner, e.g., without the need for relatively complicated polarization-de-multiplexing processing in the electrical digital domain.

In one embodiment, controller 122 generates control signal 104 using a blind polarization-equalization scheme, e.g., as described in more detail below in reference to FIGS. 2 and 3. One beneficial feature of this scheme is that it is independent of and does not rely on the phase and frequency locking implemented in the receiver's optical phase-lock loop comprising a phase detector 140, a loop filter 144, and an optical local-oscillator (OLO) source 148. In various embodiments, polarization D-MUX 106 can be implemented as a single-tap optical equalizer (see, e.g., FIG. 2) or a multi-tap optical equalizer. The additional taps in polarization D-MUX 106 can be used, e.g., to implement some optical dispersion-

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compensation processing (e.g., polarization-mode-dispersion compensation) in addition to optical polarization demultiplexing.

Each of optical signals **108_x** and **108_y** is applied to a corresponding one of optical hybrids **110_x** and **110_y**. An optical hybrid **110** has (i) two input ports labeled S and R and (ii) four output ports labeled 1 through 4. Input port S is configured to receive a corresponding one of optical signals **108_x/108_y**. Input port R is configured to receive a respective portion (e.g., an attenuated copy) of an optical reference (e.g., OLO) signal **150** generated by OLO source **148**. OLO signal **150** is phase- and frequency-locked to the carrier wave of optical input signal **102**, e.g., as described in more detail below in reference to FIGS. 4 and 5, to enable homodyne detection of signals **108_x/108_y** in receiver **100**.

In one embodiment, optical hybrid **110** splits each of the signals received at ports S and R into two (attenuated) copies, e.g., using a conventional 3-dB power splitter (not explicitly shown in FIG. 1). A relative phase shift of about 90 degrees ($\pi/2$ radian) is then applied to one copy of the signal received at port S and one copy of the signal received at port R, e.g., using phase shifters (not explicitly shown in FIG. 1). The various signal copies are then optically mixed with each other, and the resulting four mixed signals are directed to output ports 1-4, respectively. Exemplary optical hybrids that can be used to implement optical hybrids **110** in receiver **100** are described, e.g., in U.S. Patent Application Publication Nos. 2007/0297806 and 2011/0038631, both of which are incorporated herein by reference in their entirety.

The eight mixed signals generated by optical hybrids **110_x** and **110_y** are directed to an array of optical-to-electrical (O/E) converters **116₁-116₄**. In one embodiment, each O/E converter **116** comprises a pair of photo-detectors (e.g., photodiodes, not explicitly shown in FIG. 1) serially connected in a differential configuration, with each photo-detector configured to receive a corresponding one of the mixed signals received from the corresponding output port of the corresponding optical hybrid **110**. The electrical output signal generated by each pair of photo-detectors is optionally amplified to generate a corresponding one of electrical signals **120_{I_x}**, **120_{Q_x}**, **120_{I_y}**, and **120_{Q_y}**.

In one embodiment, the sampling rate of O/E converters **116₁-116₄** is controlled by a clock signal (CLK) generated by a clock-recovery circuit **130**. In one embodiment, clock-recovery circuit **130** generates clock signal CLK in a manner that causes the clock signal to have a frequency that is equal to the per-polarization symbol rate of optical input signal **102**. Representative examples of clock-recovery circuits that can be used as clock-recovery circuit **130** in receiver **100** are disclosed, e.g., in U.S. Pat. Nos. 8,290,107, 8,190,032, 7,805,083, and 7,209,666, all of which are incorporated herein by reference in their entirety.

As the signal notation implies, electrical signals **120_{I_x}** and **120_{Q_x}** represent the in-phase (I) and quadrature (Q) components, respectively, of optical signal **108_x**. Similarly, electrical signals **120_{I_y}** and **120_{Q_y}** represent the I and Q components, respectively, of optical signal **108_y**. Copies of electrical signals **120_{I_x}**, **120_{Q_x}**, **120_{I_y}**, and **120_{Q_y}** are supplied to controller **122**, which uses them to generate control signal **104**. Copies of electrical signals **120_{I_x}** and **120_{Q_y}** are also supplied to phase detector **140**, which uses them to enable phase and frequency locking of OLO signal **150** to the carrier wave of optical input signal **102**.

Each of electrical signals **120_{I_x}**, **120_{Q_x}**, **120_{I_y}**, and **120_{Q_y}** is converted into digital form in a corresponding one of analog-to-digital (A/D) slicer circuits **124₁-124₄**. Digital signals **128_{I_x}**, **128_{Q_x}**, **128_{I_y}**, and **128_{Q_y}** produced by A/D slicer cir-

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cuits **124₁-124₄** are then decoded in a conventional manner by decoders **136₁** and **136₂** to generate data streams **198_x** and **198_y**. Depending on the constellation type and forward-error-correction (FEC) coding (if any) used at the transmitter for the generation of signal **102**, decoders **136₁** and **136₂** may be configured to coordinate their decoding operations, e.g., using a communication path **138** between them. For example, U.S. patent application Ser. No. 13/537,131, entitled "FORWARD ERROR CORRECTION FOR AN OPTICAL TRANSPORT SYSTEM," discloses a four-dimensional PDM constellation and various FEC-coding schemes that, if used in the generation of signal **102**, would prompt the use of path **138** in receiver **100**. (This U.S. Patent Application is incorporated herein by reference in its entirety.) On the other hand, communication path **138** can be omitted or disabled in receiver **100** when the transmitter generates signal **102** by treating each polarization as a separate and independent communication channel.

In the embodiment shown in FIG. 1, clock-recovery circuit **130** is configured to generate clock signal CLK based on digital signal **128_{I_x}**. In an alternative embodiment, clock-recovery circuit **130** can be configured to generate clock signal CLK based on any of digital signals **128_{Q_x}**, **128_{I_y}**, and **128_{Q_y}**, or based on any suitable combination of digital signals **128_{I_x}**, **128_{Q_x}**, **128_{I_y}**, and **128_{Q_y}**.

In the embodiment shown in FIG. 1, copies of digital signals **128_{I_y}** and **128_{Q_y}** are provided to phase detector **140**, which uses them, together with electrical signals **120_{I_x}** and **120_{Q_x}**, to enable phase and frequency locking of OLO signal **150** to the carrier wave of optical input signal **102**. More specifically, an electrical phase-error signal **142** generated by phase detector **140** has a voltage proportional to the phase difference between OLO signal **150** and the carrier wave of optical input signal **102**. Loop filter **144** integrates and low-pass filters, in continuous time, phase-error signal **142** to generate a drive signal **146** for OLO source **148**. In general, frequency characteristics of loop filter **144** determine the overall dynamic time-response characteristics of the receiver's optical phase-lock loop. For example, a narrower bandwidth of loop filter **144** generally enables a more accurate phase lock, e.g., a smaller phase error. The higher accuracy however comes at the expense of a slower time response and a narrower capture range.

In an alternative embodiment, phase detector **140** can be configured to use copies of signals **128_{I_x}**, **128_{Q_x}**, **120_{I_x}**, and **120_{Q_x}** instead of signals **128_{I_y}**, **128_{Q_y}**, **120_{I_y}**, and **120_{Q_y}**, respectively.

FIG. 2 shows a block diagram of an optical polarization D-MUX **200** that can be used as optical polarization D-MUX **106** (FIG. 1) according to an embodiment of the disclosure. For illustration purposes, polarization D-MUX **200** is shown in FIG. 2 as being configured to: (i) receive optical input signal **102**; (ii) generate optical signals **108_x** and **108_y**; and (iii) receive control signal **104** having signal components labeled **104₁-104₄**. As further explained below in reference to FIG. 3, each of signal components labeled **104₁-104₄** can itself be a multi-component signal.

Optical input signal **102** is applied to a polarization beam splitter **202** configured to split this signal into two orthogonally polarized components labeled **204_x** and **204_y**. The orientation of polarization beam splitter **202** is fixed and defines the principal polarization axes (x and y) of polarization D-MUX **200**. As indicated above, polarization rotation in the optical fiber causes the orientation of the PDM components of optical input signal **102** with respect to the principal polarization axes of polarization D-MUX **200** to vary over time. As a result, each of optical signals **204_x** and **204_y** generally has

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contributions from both PDM components of optical input signal **102**. However, the optical signal processing implemented in polarization D-MUX **200** causes (i) optical signal **108_x** to primarily carry light corresponding to the first PDM component of optical input signal **102** with relatively small (e.g., negligible, for practical purposes) amounts of light corresponding to the second PDM component of the optical input signal and (ii) optical signal **108_y** to primarily carry light corresponding to the second PDM component of the optical input signal with relatively small amounts of light corresponding to the first PDM component of the optical input signal. These properties of optical signal **108_x** and **108_y** are beneficial, e.g., because they enable the subsequent decoding of these signals to be relatively straightforward and the corresponding signal-processing circuitry to be relatively inexpensive.

The optical signal processing in polarization D-MUX **200** is implemented using four optical IQ modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** arranged in a butterfly configuration, as indicated in FIG. 2. More specifically, each of IQ modulators **220₁₁** and **220₂₁** is configured to receive an attenuated copy of optical signal **204_x** generated by a polarization-maintaining coupler (PMC) **210₁**. Similarly, each of IQ modulators **220₁₂** and **220₂₂** is configured to receive an attenuated copy of optical signal **204_y** generated by a polarization-maintaining coupler **210₂**. The output signals generated by IQ modulators **220₁₁** and **220₁₂** are combined in a polarization-maintaining coupler **210₃** to generate optical signal **108_x**. The output signals generated by IQ modulators **220₂₁** and **220₂₂** are similarly combined in a polarization-maintaining coupler **210₄** to generate optical signal **108_y**.

In one embodiment, each of optical IQ modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** can be implemented using a Mach-Zehnder modulator that is commercially available from Fujitsu Optical Components Limited of Japan as part number FTM7967EQA. In an alternative embodiment, a suitable InP-based Mach-Zehnder modulator can similarly be used. In yet another alternative embodiment, each of optical IQ modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** can be implemented using the IQ modulator shown in FIG. 6.

Each polarization-maintaining coupler **210** is a three-port device that can route an optical signal between the corresponding two optically coupled ports substantially without altering the polarization state of the signal. Depending on the light propagation direction, polarization-maintaining coupler **210** can operate as an optical splitter or as an optical combiner. For example, each of polarization-maintaining couplers **210₁** and **210₂** operates as an optical splitter. In contrast, each of polarization-maintaining couplers **210₃** and **210₄** operates as an optical combiner. Representative polarization-maintaining couplers that can be used to implement polarization-maintaining couplers **210** in optical polarization D-MUX **200** are disclosed, e.g., in U.S. Pat. Nos. 6,463,195, 6,504,961, 7,203,397, and 7,373,028, all of which are incorporated herein by reference in their entirety.

Each of optical IQ modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** is configured, using the corresponding one of control signals **104₁**-**104₄**, to modulate the respective optical input signal **212** so that the resulting optical output signal **222** generated by the IQ modulator is described by Eq. (1):

$$S_{out} = H_{im} S_{in} \quad (1)$$

where S_{out} is the optical output signal; S_{in} is the optical input signal; and H_{im} is the transfer function of IQ modulator **220_{im}** ($i=1, 2$; $m=1, 2$). In general, each of S_{out} , S_{in} , and H_{im} is represented by a complex value. In one embodiment, controller **122** generates control signals **104₁**-**104₄** to cause IQ

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modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** to have transfer functions described by Eqs. (2a)-(2i):

$$H_{im} = \Gamma_{im} + j \Phi_{im} \quad (2a)$$

$$\Gamma_{11}[k] = \Gamma_{11}[k-1] + \beta I_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2) \quad (2b)$$

$$\Phi_{11}[k] = \Phi_{11}[k-1] + \beta Q_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2) \quad (2c)$$

$$\Gamma_{12}[k] = \Gamma_{12}[k-1] + \beta I_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2) \quad (2d)$$

$$\Phi_{12}[k] = \Phi_{12}[k-1] + \beta Q_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2) \quad (2e)$$

$$\Gamma_{21}[k] = \Gamma_{21}[k-1] + \beta I_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2) \quad (2f)$$

$$\Phi_{21}[k] = \Phi_{21}[k-1] + \beta Q_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2) \quad (2g)$$

$$\Gamma_{22}[k] = \Gamma_{22}[k-1] + \beta I_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2) \quad (2h)$$

$$\Phi_{22}[k] = \Phi_{22}[k-1] + \beta Q_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2) \quad (2i)$$

where Γ_{im} and Φ_{im} are the real and imaginary parts, respectively, of transfer function H_{im} (as defined in Eq. (2a)); k is a time-slot index; β is an error-weighting coefficient; I_x , Q_x , I_y , and Q_y are the magnitudes of electrical signals **120_{I_x}**, **120_{Q_x}**, **120_{I_y}**, and **120_{Q_y}** (see FIG. 1); and R is a parameter representing the average amplitude in the constellation that has been used by the transmitter to generate of signal **102**. Note that each of Eqs. (2b)-(2i) is a recursive formula that enables controller **122** to determine the value of transfer function H_{im} for the k -th time slot based on the magnitudes of electrical signals **120_{I_x}**, **120_{Q_x}**, **120_{I_y}**, and **120_{Q_y}** and the value of transfer function H_{im} in the previous (i.e., $(k-1)$ -th) time slot. Further note that the second term in the right part of each of Eqs. (2b)-(2i) provides a measure of the misalignment between the current state of polarization of signal **102** and the orientation of the principal polarization axes of polarization D-MUX **200**.

When the operative constellation is a phase-shift keying (PSK) constellation, all constellation points are located on a circle centered at the origin of the constellation (e.g., the origin of the corresponding complex plane). In this case, parameter R represents the radius of this circle. When the operative constellation is a quadrature-amplitude modulation (QAM) constellation, different constellation points may have different amplitudes, which is manifested by their respective different distances from the origin of the constellation. In this case, parameter R can be calculated, e.g., using Eq. (3):

$$R = \frac{\alpha}{M} \sum_{i=1}^M r_i \quad (3)$$

where M is the number of constellation points in the QAM constellation; α is a scaling coefficient that reflects signal amplification/attenuation between the transmitter and receiver; and r_i is the normalized amplitude of the i -th constellation point in the QAM constellation. Note that Eq. (3) is also valid for a PSK constellation, wherein all r_i values are the same.

Suppose now that IQ modulators **220₁₁**, **220₂₁**, **220₁₂**, and **220₂₂** are configured so that the polarization rotation imposed on input signal **102** in the optical transport link is fully reversed, and signals **108_x** and **108_y** represent the "pure" first and second PDM components, respectively, of the input signal. Then, due to the way parameter R is defined (see, e.g., Eq. (3)), the time average of each of the second terms in Eqs. (2b)-(2i) is zero, and the configurations of IQ modulators

220_{11} , 220_{21} , 220_{12} , and 220_{22} manifested by their respective transfer functions (H_{im}) remain substantially unchanged over time. However, when the polarization rotation imposed on input signal **102** in the optical transport link is not fully reversed in polarization D-MUX **200**, the second terms in Eqs. (2b)-(2i) do not average to zero, thereby causing controller **122** to appropriately adjust transfer functions H_{im} . This adjustment then causes polarization D-MUX **200** to more accurately undo the polarization rotation and to track its changes over time.

Error-weighting coefficient β is an empirical parameter whose value is set to be relatively small to ensure smooth polarization-rotation tracking by polarization D-MUX **200**. For example, the relatively small value of error-weighting coefficient β enables sufficient time averaging to occur for proper polarization-rotation tracking when input signal **102** is an M-QAM signal. The relatively small value of error-weighting coefficient β also enables polarization D-MUX **200** not to overreact to burst errors, which ensures relatively slow and gradual adjustment of the overall polarization D-MUX configuration.

FIGS. 3A-3B show block diagrams of a controller **300** that can be used as controller **122** (FIG. 1) according to an embodiment of the disclosure. More specifically, FIG. 3A shows an overall block diagram of controller **300**. FIG. 3B shows a block diagram of a filter **390**, eight instances (copies) of which are used in the circuit shown in FIG. 3A. Controller **300** is an electrical controller that implemented using various analog circuits as its building blocks. When controller **300** is used in receiver **100** configured to operate under a relatively low-bit QAM format, such as 16-QAM or 32-QAM, the linearity requirements to the analog circuits used in the controller are not very stringent. For example, controller **300** can be implemented using analog circuits that have a relatively small linear dynamic range.

Controller **300** comprises a signal generator **306**, five squaring circuits 310_1 - 310_5 , two three-input signal adders 320_1 - 320_2 , eight analog signal multipliers **330**, and eight filters 390_1 - 390_8 , all interconnected as indicated in FIG. 3A. Each filter **390** comprises a signal generator **326**, an analog signal multiplier **330**, a delay element (Z^{-1}) **350**, and a two-input signal adder **370**, all interconnected as indicated in FIG. 3B.

Signal generator **306** has a register (not explicitly shown in FIG. 3A) that stores the value of parameter R (see Eq. (3)). An output signal **308** generated by signal generator **306** is an electrical analog signal having a magnitude proportional to the parameter value stored in the register.

Squaring circuit 310_i is configured to generate an output signal 312_i that is proportional to the square of the magnitude of a received input signal, where $i=1, 2, \dots, 5$. As indicated in FIG. 3A, the input signals received by squaring circuits 310_1 - 310_5 are electrical signals 120_{Ix} , 120_{Qx} , **308**, 120_{Iy} , and 120_{Qy} , respectively.

Signal adder 320_1 is configured to add signals 312_1 and 312_2 , and subtract from the sum signal 312_3 . The resulting signal generated by signal adder 320_1 is signal 322_1 .

Signal adder 320_2 is similarly configured to add signals 312_4 and 312_5 , and subtract from the sum signal 312_3 . The resulting signal generated by signal adder 320_2 is signal 322_2 .

The eight analog signal multipliers **330** are configured to variously scale signals 322_1 and 322_2 , using signals 120_{Ix} , 120_{Qx} , 120_{Iy} , and 120_{Qy} as scaling factors. The resulting scaled signals generated by the eight analog signal multipliers **330** are signals 334_1 - 334_8 . The magnitudes of signals 334_1 - 334_8 are proportional to the following quantities:

$$\text{signal } 334_1: I_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2);$$

$$\text{signal } 334_2: Q_x[k]((I_x[k])^2 + (Q_x[k])^2 - R^2);$$

$$\text{signal } 334_3: I_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2);$$

$$\text{signal } 334_4: Q_y[k]((I_y[k])^2 + (Q_y[k])^2 - R^2);$$

$$\text{signal } 334_5: I_x[k]((I_y[k])^2 + (Q_y[k])^2 - R^2);$$

$$\text{signal } 334_6: Q_x[k]((I_y[k])^2 + (Q_y[k])^2 - R^2);$$

$$\text{signal } 334_7: I_y[k]((I_x[k])^2 + (Q_x[k])^2 - R^2); \text{ and}$$

$$\text{signal } 334_8: Q_y[k]((I_x[k])^2 + (Q_x[k])^2 - R^2).$$

Filters 390_1 - 390_8 are configured to implement signal processing corresponding to Eqs. (2b)-(2i), respectively. More specifically, signal generator **326** and the analog signal multiplier **330** in filter 390_i are configured to scale signal 334_i using error-weighting coefficient β as a scaling factor. A resulting scaled signal **368** is then used for the recursive update of an output signal 392_i , with delay element **350** providing an appropriate time delay for said recursive update. As a result, output signals 390_1 - 390_8 generated by filters 390_1 - 390_8 are proportional to $\Gamma_{11}[k]$, $\Phi_{11}[k]$, $\Gamma_{12}[k]$, $\Phi_{12}[k]$, $\Gamma_{21}[k]$, $\Phi_{21}[k]$, $\Gamma_{22}[k]$, and $\Phi_{22}[k]$, respectively (also see Eqs. (2b)-(2i)).

Signals 390_1 and 390_2 provide the I and Q components, respectively, for control signal 104_1 (FIG. 2). Signals 390_3 and 390_4 provide the I and Q components, respectively, for control signal 104_2 (FIG. 2). Signals 390_5 and 390_6 provide the I and Q components, respectively, for control signal 104_3 (FIG. 2). Signals 390_7 and 390_8 provide the I and Q components, respectively, for control signal 104_4 (FIG. 2).

FIG. 4 shows a block diagram of a phase detector **400** that can be used as phase detector **140** (FIG. 1) according to an embodiment of the disclosure. For illustration purposes, phase detector **400** is shown in FIG. 4 as being configured to (i) receive electrical signals 120_{Iy} and 120_{Qy} , and digital signals 128_{Iy} and 128_{Qy} , and (ii) generate phase-error signal **142**. Other signal configurations for phase detector **400** are also possible. For example, phase detector **400** can alternatively be configured to use signals 128_{Ix} , 128_{Qx} , 120_{Ix} , and 120_{Qx} instead of signals 128_{Iy} , 128_{Qy} , 120_{Iy} , and 120_{Qy} , respectively.

Phase detector **400** comprises multipliers **402** and **404** and an adder **410** configured to implement the signal-processing operations expressed by Eq. (4):

$$\epsilon = I'_y Q_y - Q'_y I_y \quad (4)$$

where ϵ is the magnitude of phase-error signal **142**; and I'_y , Q'_y , Q_y , and I_y are the magnitudes of electrical signals 128_{Iy} , 120_{Qy} , 128_{Qy} , and 120_{Iy} , respectively (see FIG. 1). Eq. (4) is based on the following property of complex values: when first and second complex values have the same argument, the product of the first complex value and the complex conjugate of the second complex value is a real number. The right side of Eq. (4) expresses the imaginary part of the product of a received symbol (as represented by electrical signals 120_{Iy} and 120_{Qy}) and the constellation point to which the received symbol is mapped after being processed by A/D slicer circuits 124_3 - 124_4 (with said constellation symbol represented by digital signals 128_{Iy} and 128_{Qy}). When the phase-lock error is relatively small, the value of ϵ is approximately proportional to the difference between the arguments of the received symbol and the corresponding constellation point. Thus, phase-error signal **142** generated by phase detector **400** as indicated in FIG. 4 provides a measure of the phase-lock error between

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OLO signal **150** and the carrier wave of optical input signal **102** (also see FIG. 1). To achieve a phase lock between OLO signal **150** and the carrier wave of optical input signal **102**, receiver **100** can then drive OLO source **148**, using phase-error signal **142** generated by phase detector **400**, in a manner that tends to minimize the absolute value of ϵ .

In one embodiment, each of multipliers **402** and **404** can be implemented using a Gilbert cell. In an alternative embodiment, each of multipliers **402** and **404** can be implemented using an XOR gate.

FIG. 5 shows a block diagram of a light source **500** that can be used as OLO source **148** (FIG. 1) according to one embodiment of the disclosure. For illustration purposes, light source **500** is shown in FIG. 5 as being configured to receive filtered phase-error signal **146** and to generate OLO signal **150**.

Light source **500** comprises a laser (e.g., a laser diode) **502** optically coupled to an optical IQ modulator **506**. On optical beam **504** generated by laser **502** nominally has the same frequency (wavelength) as the carrier wave of optical input signal **102**. However, the phase noise, e.g., due to the finite linewidth in optical beam **504**, generally causes a non-zero frequency and phase offset between optical beam **504** and the carrier wave of optical input signal **102**. Optical IQ modulator **506** is configured to modulate optical beam **504** so as to remove the frequency and phase offset and achieve a requisite phase lock. A modulated optical signal **508** generated by IQ modulator **506** is then filtered using an optical filter **510** to generate OLO signal **150**. In one embodiment, optical IQ modulator **506** can be implemented using a Mach-Zehnder modulator, such as the above-mentioned model FTM7967EQA manufactured by Fujitsu Optical Components Limited of Japan. In an alternative embodiment, optical IQ modulator **506** can be implemented using the IQ modulator shown in FIG. 6.

Electrical drive signals **512**₁ and **512**₂ for optical IQ modulator **506** are generated using a voltage-controlled oscillator (VCO) **520** and amplifiers **516**₁ and **516**₂. Voltage-controlled oscillator **520** is an electronic circuit configured to generate electrical RF signals **518**₁ and **518**₂ so that (i) these two RF signals have the same frequency proportional to the magnitude (e.g., quasi-dc voltage) of filtered phase-error signal **146** generated by loop filter **144**, e.g., by filtering phase-error signal **142** generated by phase detector **400**, and (ii) there is a fixed phase offset of approximately 90 degrees between these two RF signals. Amplifiers **510**₁ and **510**₂ are configured to appropriately amplify and bias electrical RF signals **516**₁ and **516**₂, thereby generating electrical drive signals **508**₁ and **508**₂ in a manner that causes optical IQ modulator **506** to apply sideband- and carrier-suppression modulation to optical beam **504**. As a result, most of the optical energy of modulated optical signal **508** generated by IQ modulator **506** is contained in a single modulation sideband. Optical filter **510** passes this relatively strong modulation sideband while rejecting or blocking the residual carrier and the other relatively weak modulation sideband. The output of optical filter **510** can then be used as OLO signal **150** in receiver **100** (FIG. 1).

In some embodiments, optical filter **510** is optional and can be removed from light source **500**.

FIG. 6 shows a block diagram of an IQ modulator **600** that can be used as modulator **220** (FIG. 2) according to an embodiment of the disclosure. When used as modulator **220**, in optical polarization D-MUX **200** (FIG. 2), IQ modulator **600** is configured to: (i) receive, as an optical input signal, optical signal **212**_i; (ii) generate optical output signal **222**_i; and (iii) be driven using control signal **104**_i.

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IQ modulator **600** comprises a nested Mach-Zehnder modulator (MZM) **610**. As used herein, the term “nested” means that each of MZM arms **612**₁ and **612**₂ of Mach-Zehnder modulator **610** itself includes, as a constituent part, a respective conventional Mach-Zehnder modulator. More specifically, MZM arm **612**₁ includes a conventional Mach-Zehnder modulator **616**₁, and MZM arm **612**₂ includes a conventional Mach-Zehnder modulator **616**₂. Each of Mach-Zehnder modulators **616**₁ and **616**₂ is shown in FIG. 6 as a rectangle, with the arms of the modulator not being explicitly shown to make FIG. 6 easier to read.

Mach-Zehnder modulator **616**₁ is driven using the I-component of control signal **104**_i. For example, each of signals **390**₁, **390**₃, **390**₅, and **390**₇ generated by controller **300** (FIG. 3) is an I-component of the corresponding control signal **104**_i.

Mach-Zehnder modulator **616**₂ is similarly driven using the Q-component of control signal **104**_i. For example, each of signals **390**₂, **390**₄, **390**₆, and **390**₈ generated by controller **300** (FIG. 3) is a Q-component of the corresponding control signal **104**_i.

Amplifiers **604**₁ and **604**₂ are configured to appropriately amplify and apply a dc bias to the I-component and the Q-component, respectively, of control signal **104**_i to generate electrical drive signals **606**₁ and **606**₂ for Mach-Zehnder modulators **616**₁ and **616**₂, respectively.

MZM arm **612**₂ further includes a fixed phase shifter **620** configured to apply a phase shift of about 90 degrees to the optical signal propagating through that MZM arm. This phase shift enables MZM arms **612**₁ and **612**₂ to operate as an I-modulation arm and a Q-modulation arm, respectively, of IQ modulator **600**.

Each of MZM arms **612**₁ and **612**₂ is illustratively shown as further including a respective one of optional configurable phase shifters **618**₁ and **618**₂. These phase shifters can be used, e.g., to compensate for possible unwanted phase imbalances between MZM arms **612**₁ and **612**₂. If no such unbalances are present in IQ modulator **600**, then each of phase shifters **618**₁ and **618**₂ can be configured to receive a zero (e.g., ground) bias potential.

While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense.

Various modifications of the described embodiments, as well as other embodiments of the invention, which are apparent to persons skilled in the art to which the invention pertains are deemed to lie within the principle and scope of the invention as expressed in the following claims.

Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value of the value or range.

The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims during the examination. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments indicated by the used figure numbers and/or figure reference labels.

It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of this invention may be made by those skilled in the art without departing from the scope of the invention as expressed in the following claims.

Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be

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included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. 5 The same applies to the term “implementation.”

Also for purposes of this description, the terms “couple,” “coupling,” “coupled,” “connect,” “connecting,” or “connected” refer to any manner known in the art or later developed in which energy is allowed to be transferred between two or more elements, and the interposition of one or more additional elements is contemplated, although not required. Conversely, the terms “directly coupled,” “directly connected,” etc., imply the absence of such additional elements.

The description and drawings merely illustrate the principles of the invention. It will thus be appreciated that those of ordinary skill in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the invention and are included within its spirit and scope. Furthermore, all examples recited herein are principally intended expressly to be only for pedagogical purposes to aid the reader in understanding the principles of the invention and the concepts contributed by the inventor(s) to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting embodiments of the invention, as well as specific examples thereof, are intended to encompass equivalents thereof.

The functions of the various elements shown in the figures, including any functional blocks labeled as “processors,” may be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions may be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which may be shared. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and may implicitly include, without limitation, digital signal processor (DSP) hardware, network processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), read only memory (ROM) for storing software, random access memory (RAM), and non volatile storage. Other hardware, conventional and/or custom, may also be included. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the implementer as more specifically understood from the context.

What is claimed is:

1. An apparatus comprising:

an optical hybrid configured to optically mix an optical input signal and an optical local-oscillator, “OLO,” signal to generate a plurality of mixed optical signals; 55
a first optical-to-electrical converter configured to convert a first subset of the plurality of mixed optical signals into a first electrical analog signal;
a second optical-to-electrical converter configured to convert a second subset of the plurality of mixed optical signals into a second electrical analog signal;
a first slicer circuit configured to slice the first electrical analog signal to generate a first electrical digital signal;
a second slicer circuit configured to slice the second electrical analog signal to generate a second electrical digital signal; and

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a phase detector configured to generate an electrical measure of a phase difference between the OLO signal and a carrier wave of the optical input signal, with said electrical measure being generated using the first electrical analog signal, the first electrical digital signal, the second electrical analog signal, and the second electrical digital signal.

2. The apparatus of claim 1, further comprising a decoder configured to recover data carried by the optical input signal from the first electrical digital signal and the second electrical digital signal.

3. The apparatus of claim 1, wherein the apparatus is configured to use said electrical measure to phase-lock the OLO signal to the carrier wave of the optical input signal.

4. The apparatus of claim 1, wherein said electrical measure is proportional to the following quantity:

$$I_y Q_y - Q_y I_y,$$

where I_y , Q_y , Q_y , and I_y are magnitudes of the first electrical digital signal, the second electrical analog signal, the second electrical digital signal, and the first electrical analog signal, respectively.

5. The apparatus of claim 1, wherein the phase detector comprises:

a first signal multiplier configured to multiply the first electrical analog signal and the first electrical digital signal to generate a first multiplied electrical signal;

a second signal multiplier configured to multiply the second electrical analog signal and the second electrical digital signal to generate a second multiplied electrical signal; and

a signal adder configured to subtract the second multiplied electrical signal from the first multiplied electrical signal to generate the electrical measure.

6. The apparatus of claim 1, further comprising:

a laser configured to generate an optical beam; and
an optical IQ modulator configured to optically modulate said optical beam to generate a modulated optical beam, wherein the apparatus is configured to generate the OLO signal using said modulated optical beam.

7. The apparatus of claim 6, wherein the optical IQ modulator is configured to optically modulate the optical beam using carrier-suppression and sideband-suppression modulation in a manner that causes most of optical energy of the modulated optical beam to be contained in a single modulation sideband.

8. The apparatus of claim 6, further comprising an optical filter configured to filter the modulated optical beam to generate the OLO signal.

9. The apparatus of claim 8, wherein:

the optical filter is configured to pass a first modulation sideband of the modulated optical beam; and
the OLO signal comprises light of the first modulation sideband that passed through the optical filter.

10. The apparatus of claim 9, the optical filter is further configured to block a carrier wave and a second modulation sideband of the modulated optical beam.

11. The apparatus of claim 6, further comprising a phase-lock loop configured to drive the optical IQ modulator using the electrical measure.

12. The apparatus of claim 11, wherein the phase-lock loop comprises:

an electrical loop filter configured to filter the electrical measure to generate a filtered measure; and

an electrical voltage-controlled oscillator configured to drive the optical IQ modulator using the filtered measure.

13. The apparatus of claim **12**,
 wherein the electrical voltage-controlled oscillator is configured to:
 generate a first electrical radio-frequency signal using
 the filtered measure; and
 generate a second electrical radio-frequency signal
 using the filtered measure, wherein the first and second
 electrical radio-frequency signals have a common radio-frequency
 proportional to the filtered measure, and there is a fixed phase
 offset of approximately 90 degrees between the first electrical
 radio-frequency signal and the second electrical radio-frequency
 signal; and
 wherein the phase-lock loop is configured to drive the
 optical IQ modulator using the first and second electrical
 radio-frequency signals.

14. The apparatus of claim **6**, wherein the IQ modulator
 comprises a nested Mach-Zehnder modulator.

15. The apparatus of claim **1**, further comprising an optical
 polarization de-multiplexer configured to de-multiplex an
 optical polarization-division-multiplexed signal into a first
 optical component having a first polarization and a second
 optical component having a second polarization orthogonal
 to the first polarization, wherein the optical input signal is one
 of said first and second optical components.

16. The apparatus of claim **15**, further comprising a controller
 configured to generate one or more control signals for
 the optical polarization de-multiplexer using the first electrical
 analog signal and the second electrical analog signal.

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